

MOHAMED SULAIMAN

Hardware Design Engineer | Embedded Electronics | PCB Design | Product R&D;

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PROFESSIONAL SUMMARY

Electronics and Communication Engineer with hands-on experience in **electronics hardware development, PCB prototyping, embedded systems, IoT product development and rapid prototyping**. Experienced in **KiCad PCB development, component selection, BOM preparation, sensor integration, hardware bring-up, board-level debugging, power management, SMD soldering and functional testing**. Currently developing real-world products using **ESP32, STM32 and Nordic microcontrollers**, combining electronics hardware with Embedded C/C++ firmware. Experienced in integrating sensors, displays, wireless modules, batteries, motor/actuator interfaces and 3D-printed mechanical components into functional prototypes.

CORE TECHNICAL SKILLS

Hardware & Electronics: PCB Design & Prototyping, KiCad, Schematic-Level Design, Component Selection, BOM Preparation, Hardware Integration, Board Bring-Up, Board-Level Debugging, Power Management, Battery-Powered Electronics, Sensor Integration, SMD Soldering, Functional Testing, Prototype Assembly

Embedded Systems: ESP32, ESP32-S3, STM32, Nordic nRF54L15, Arduino, Embedded C/C++, FreeRTOS Concepts, Zephyr RTOS, Firmware Debugging

Interfaces & Connectivity: I2C, SPI, UART, GPIO, PWM, ADC, BLE, Wi-Fi, MQTT, ESP-NOW

Sensors & Electronics: Load Cells, IMUs, Motion Sensors, TFT/OLED Displays, Sensor Interfaces, Motor Drivers, Servo Motors, Stepper Motors, Actuator Interfaces

Rapid Prototyping & Mechanical: Fusion 360, FDM 3D Printing, Ender 3, Bambu Lab, Klipper, Mechanical Parts, Enclosures, Brackets, Fixtures, Mechanical-Electronic Integration

Development Tools: PlatformIO, Arduino IDE, ESP-IDF, nRF Connect SDK, VS Code, Git, GitHub, EEZ Studio, Python

PROFESSIONAL EXPERIENCE

Cavin Infotech — IoT Junior Engineer / R&D;

Chennai, Tamil Nadu | Oct 2025–Present

- Develop and test embedded IoT product prototypes using **ESP32, STM32 and Nordic-based microcontrollers**.
- Integrate **sensors, displays, wireless modules, batteries and electronic components** into functional product hardware.
- Perform **hardware bring-up, board-level debugging, functional testing and iterative hardware improvements** during product development.
- Develop Embedded C/C++ firmware using **BLE, Wi-Fi, I2C, SPI and UART** for connected and battery-powered devices.
- Troubleshoot hardware, firmware, communication and sensor-integration issues during prototype development.
- Developed Smart Collar functionality involving **BLE communication, time synchronization and battery-life improvements**.
- Integrated a larger **TFT color display** and optimized embedded UI for a connected smart-product prototype.
- Collaborate across hardware, firmware and product-development activities to convert requirements into working prototypes.

Tarcin Robotic LLP — Product Developer / R&D;

Tamil Nadu | 7 months

- Developed IoT and embedded product prototypes using **ESP32 and STM32** for robotics and product-development applications.
- Integrated **sensors, actuators, displays and communication modules** into functional prototype systems.
- Designed and fabricated mechanical prototypes using **Fusion 360 and FDM 3D printing**.
- Performed hardware assembly, firmware development, functional testing and troubleshooting.
- Supported rapid product development through iterative **build-test-debug** cycles.

Rasoi Robotics Pvt. Ltd. — R&D; Intern

Tamil Nadu | 5 months

- Worked on automation and robotics R&D; projects involving **embedded controllers, sensors and data acquisition**.
- Performed sensor integration, hardware testing and troubleshooting of hardware-software integration issues.
- Assisted with **PCB development and testing** during prototype development.

Hashind Solutions — Part-Time Project Engineer

India | 1 year

- Built embedded and robotics projects integrating **microcontrollers, sensors, wireless communication and automation**.
- Developed practical robotics and embedded-system prototypes for educational and project applications.
- Trained **2,000+ students** in robotics and embedded systems through hands-on project development.
- Supported WRO teams, with **2 teams reaching state finals and 1 advancing to an international round**.

SELECTED HARDWARE & EMBEDDED PROJECTS

SmartBowl — Connected Product Prototype

ESP32-S3 | Sensors | TFT Display | Battery | SPI | I2C | 3D Printing

- Built a connected product prototype integrating **embedded electronics, sensors, display and wireless communication**.
- Integrated a larger TFT display, optimized embedded UI and performed hardware, firmware and sensor debugging.

- Designed and iterated physical prototype components using **3D printing**.

Smart Pet Activity Tracking Collar

nRF54L15 / ESP32 | BLE | Sensors | Battery | Fusion360

- Developed a wearable connected prototype for pet activity monitoring.
- Implemented BLE communication, synchronization and event-based firmware functionality.
- Integrated sensors into a compact battery-powered product and worked on power optimization.
- Designed the mechanical enclosure using Fusion 360.

ESP32 Drawing Robot

ESP32 | StepperMotors | Servo | MotorDriver

- Developed a mobile robotic platform using ESP32, stepper motors and servo-based pen control.
- Integrated motor drivers and developed motion-control firmware.
- Diagnosed motor direction, vibration and movement problems through iterative hardware, firmware and mechanical testing.

Multi-Sensor Embedded Monitoring System

ESP32-S3 | MPU6050 | GY-87 | MCP23017 | TFT | I2C | SPI

- Integrated multiple sensors and peripherals with display and control interfaces for a working embedded prototype.
- Unified peripheral drivers into a reusable HAL-style firmware structure.
- Implemented sensor acquisition, event detection and RTOS-style task separation.

IoT Data Acquisition & Dashboard

ESP32 | MQTT | Node-RED | InfluxDB | Grafana

- Developed an ESP32-based data-acquisition system integrating sensors and real-time MQTT communication.
- Performed end-to-end hardware, firmware and communication testing.
- Built real-time data visualization using InfluxDB and Grafana.

PCB & ELECTRONICS DEVELOPMENT

- PCB design and prototyping using **KiCad**.
- Schematic-level electronics development, **component selection and BOM preparation**.
- Experience with ESP32-based custom electronics, power regulation, sensor interfaces and displays.
- Hands-on experience with **load cells, ADCs, IMUs, TFT/OLED displays, motor/servo interfaces, SMD components and power-management circuits**.
- Hardware bring-up, board-level debugging, prototype assembly and functional validation.

3D DESIGN & RAPID PROTOTYPING

- 5+ years of hands-on **FDM 3D printing** experience with Ender 3 and Bambu Lab platforms.
- Fusion 360 design of functional parts, enclosures, brackets, fixtures and prototype mechanisms.
- Klipper configuration and 3D-printer tuning.
- Iterative product development: **design → print → assemble → test → modify → reprint**.
- Mechanical-electronic integration for functional embedded product prototypes.

EDUCATION

Bachelor of Engineering (B.E.) — Electronics & Communication Engineering

SSM Institute of Engineering and Technology, Dindigul | 2021 – 2025

ACHIEVEMENTS

- **Circle of Excellence Award** — Cavin Infotech / CavinKare R&D; Center.
- Trained **2,000+ students** in robotics and embedded systems.
- Supported WRO teams with **2 state-finalists and 1 international-round advancement**.
- Won **₹10,000** for innovation in safety technology.

LANGUAGES

English | Tamil

TARGET ROLES

Hardware Design Engineer | Electronics Design Engineer | Embedded Hardware Engineer | Hardware R&D; Engineer | Electronics R&D; Engineer | Embedded Systems Engineer | Product Development Engineer